插图英语阅读小册

第3册 · 十篇故事

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让蚂蚁变僵尸的真菌

The Fungus That Turns Ants into Zombies



蚂蚁为何会咬住叶片不松口?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Deep in a Thai rainforest, a carpenter ant breaks from its colony's column in the lofty canopy. Swaying on unsteady legs, it wanders downward, as if intoxicated, into leafy undergrowth. Around midday it clenches its jaws on a major vein underneath a leaf, about 25 centimetres above the ground.

By dusk the insect is lifeless, yet the bite holds rigidly. A fungus has destroyed its jaw muscles, so they cannot loosen. Two or three days later a fungal stalk pushes from its head and releases infectious spores on to the ground below, where foraging ants tread.

The invader, Ophiocordyceps unilateralis, is fussy about location. David Hughes and colleagues found the bites mostly on the north-west side of plants, where the air is humid and the temperature suits the fungus. When researchers moved infected ants higher, the fungus grew abnormally. Placed on the ground, the bodies disappeared.

In 2017 a Penn State team examined infected ants with three-dimensional microscope imaging. Fungal cells had formed an intertwined lattice around muscle fibres in the legs and jaw, a picture that evokes the strings of a puppet. Yet the fungus had not infiltrated the brain.

A Planet Earth episode, with David Attenborough's narration, showed one such ant. It helped inspire The Last of Us, a 2013 video game whose makers envisioned the parasite jumping to humans. The hypothetical scenario gave gamers a fictional nightmare.

The real parasite's roots run far deeper, into the geological record. A 48-million-year-old fossil leaf from Messel, in Germany, bears the same characteristic bite scars.

中文译文

在泰国雨林深处，一只木匠蚁脱离了高处树冠中的蚁群队列。它腿脚不稳，摇摇晃晃地向下游荡，仿佛喝醉一般，钻进枝叶茂密的林下植被。正午前后，它用双颚紧紧咬住一片叶子背面的主叶脉，那里离地面约25厘米。

到黄昏时，蚂蚁已经死去，却仍牢牢咬住叶子。真菌破坏了它的颚部肌肉，使其无法放松。两三天后，一根菌柄从蚂蚁头部伸出，向下方地面散播具有传染性的孢子，那里正有觅食的蚂蚁经过。

这种入侵者叫偏侧蛇虫草菌，对位置很挑剔。戴维·休斯及其同事发现，咬合点大多位于植物的西北侧，那里空气潮湿，温度也适合真菌。研究者将受感染的蚂蚁移到更高处后，真菌生长得不正常；放到地面的蚁尸则不见了。

2017年，宾夕法尼亚州立大学的团队通过三维显微成像检查受感染的蚂蚁。真菌细胞在腿部和颚部的肌肉纤维周围织成相互交错的网络，令人联想到操纵木偶的提线。然而，真菌并未侵入蚂蚁的大脑。

由戴维·阿滕伯勒解说的纪录片《地球脉动》曾在一集中展示过这样一只蚂蚁。这启发了2013年电子游戏《最后生还者》的创作者，他们设想这种寄生菌转而感染人类。这种假想情景给玩家带来了一场虚构的噩梦。

现实中，这种寄生现象的历史要久远得多，可以追溯到地质记录之中。德国梅塞尔出土的一片4800万年前的叶子化石上，留有同样特征的咬痕。

核心词汇

先记这22项。连着下方的原文短语一起回想。

colony n.
（蚂蚁等的）群体
its colony's column

canopy n.
森林的树冠层
in the lofty canopy

unsteady adj.
不稳的
on unsteady legs

wander v.
游荡；漫无目的地走
it wanders downward

intoxicated adj.
醉酒般的
as if intoxicated

undergrowth n.
林下灌丛；林下植被
into leafy undergrowth

clench v.
紧咬；紧闭
it clenches its jaws

vein n.
叶脉
a major vein underneath a leaf

dusk n.
黄昏
By dusk

rigidly adv.
僵硬地；牢牢地
the bite holds rigidly

infectious adj.
具有传染性的
releases infectious spores

tread v.
行走；踏过
where foraging ants tread

fussy adj.
挑剔的
is fussy about location

humid adj.
潮湿的
the air is humid

abnormally adv.
不正常地；反常地
the fungus grew abnormally

intertwined adj.
交织的；缠绕的
an intertwined lattice

lattice n.
网格；格状结构
an intertwined lattice

evoke v.
使人联想到
a picture that evokes

infiltrate v.
侵入；渗入
had not infiltrated the brain

hypothetical adj.
假想的；假设的
The hypothetical scenario

scenario n.
情景；设想
The hypothetical scenario

fossil n.
化石
fossil leaf from Messel

文中词汇与搭配

补充词汇共31项。保留每个表达在本文中的意思。

rainforest n.
雨林

carpenter n.
木匠（木匠蚁名称的一部分）

column n.
行进的队列

lofty adj.
高耸的；高处的

sway v.
摇晃；摇摆

downward adv.
向下

as if 连词短语
仿佛；好像

leafy adj.
枝叶茂密的

midday n.
正午

jaw n.
颚；口部咬合器官

lifeless adj.
死去的；无生命的

muscle n.
肌肉

loosen v.
放松；松开

fungal adj.
真菌的

invader n.
入侵者

location n.
位置；地点

bite n.
咬合点；咬痕

north-west adj.
西北侧的；西北方的

disappear v.
消失；不见

three-dimensional adj.
三维的

microscope n.
显微镜

puppet n.
木偶

narration n.
解说；旁白

inspire v.
启发；赋予灵感

maker n.
创作者；制作者

envision v.
设想；想象

gamer n.
电子游戏玩家

fictional adj.
虚构的

geological adj.
地质的

characteristic adj.
有代表性的；典型的

scar n.
疤痕；留下的痕迹

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Around midday it _____ its jaws on a major vein underneath a leaf, about 25 centimetres above the ground.

02 When researchers moved infected ants higher, the fungus grew _____.

03 Yet the fungus had not _____ the brain.

04 The _____ scenario gave gamers a fictional nightmare.

文章主线

- 感染 泰国木匠蚁感染偏侧蛇虫草菌后离群，正午咬住离地约25厘米的叶背主脉，黄昏死去。
- 传播 真菌破坏颚肌，使蚂蚁死后仍咬紧；两三天后菌柄顶出，孢子散向下方蚂蚁觅食的地面。
- 选址 咬痕多在温湿度适合的植物西北侧；感染蚂蚁移高后真菌生长反常，放到地面的蚁尸则消失。
- 成像 2017年三维成像发现，真菌在腿、颚肌肉周围织成网络，却没有侵入蚂蚁大脑。
- 边界 4800万年前的叶化石已有同类咬痕；纪录片启发了游戏《最后生还者》，但感染人类的情节是虚构。

参考答案　1 clenches 2 abnormally 3 infiltrated 4 hypothetical

禁区里的生机

Life in the Forbidden Zone



禁区里动物变多，就能说明辐射无害吗?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Chernobyl is synonymous with disaster, yet it holds a startling amount of wildlife. Early on 26 April 1986, one of its reactors exploded in Soviet Ukraine, dispersing radioactive fallout across Europe. On 27 April, nearby Pripyat was emptied of about 45,000 inhabitants within hours. A zone some 30 kilometres in radius became off-limits, and Belarus fenced off adjoining terrain.

In October 2015, a study in Current Biology combined aerial surveys with footprint counts on the Belarusian side. Elk, roe deer, red deer, and wild boar were comparable in density to those in four reserves free of contamination, and wolves were over seven times as numerous. Earlier helicopter counts showed elk, roe deer, and boar surging in the post-disaster decade.

Later fieldwork bolstered these findings. In April 2016, camera traps snapped wolves, boar, and raccoon dogs near the most hazardous spots, with no sign of suppressed numbers. The probable reason is mundane: the absence of agriculture, forestry, and hunting.

Smaller organisms fared worse in some studies. One, in March 2007, found fewer nesting birds, and fewer species, where radiation was more intense. Another, in March 2009, found pollinators and spiders less plentiful where exposure was greater. A third, in March 2014, found leaves decomposing about 40 per cent less at the most polluted sites, impeding the return of nutrients to the soil.

Counting heads cannot guarantee the health of individual animals, and how much damage low but continual doses inflict remains contentious, as studies clash over methods. The burgeoning herds may trumpet the dividend of human departure, but the sluggish rot of fallen leaves keeps a quieter ledger of radiation's cost.

中文译文

切尔诺贝利是灾难的代名词，却栖息着数量惊人的野生动物。1986年4月26日凌晨，苏联乌克兰境内这里的一座反应堆爆炸，放射性尘降物散布到欧洲各地。4月27日，附近普里皮亚季的约4.5万居民在数小时内撤离。半径约30公里的区域被列为禁区，白俄罗斯也将相邻地带围封起来。

2015年10月，《当代生物学》发表的一项研究结合白俄罗斯一侧的空中调查与足迹计数。当地驼鹿、狍、马鹿和野猪的密度，与四个未受污染保护区的同类相当；狼的数量则为对照保护区的七倍多。较早的直升机计数显示，灾后十年里，驼鹿、狍和野猪数量激增。

后来的实地研究为这些发现提供了支持。2016年4月，自动感应相机在危险程度最高的一些地点附近拍到狼、野猪和貉，没有发现数量受抑制的迹象。可能的原因其实很普通：农业、林业和狩猎活动消失了。

但一些研究发现，较小的生物境况更差。2007年3月的一项研究发现，辐射较强的地方，筑巢鸟类的数量和鸟种都较少。2009年3月的另一项研究发现，辐射暴露更高的地方，传粉动物和蜘蛛较少。2014年3月的第三项研究发现，污染最严重地点的落叶分解量少了约40%，妨碍养分回到土壤。

清点动物数量，不能保证每一个体都健康。持续低剂量辐射究竟造成多大伤害，仍有争议，各项研究在方法上也存在分歧。不断扩大的兽群，或许彰显了人类离开带来的好处；而落叶迟缓的腐烂，则默默记下了辐射的另一笔代价。

核心词汇

先记这22项。连着下方的原文短语一起回想。

synonymous with 形容词短语
是……的代名词
is synonymous with disaster

disperse v.
散布；扩散
dispersing radioactive fallout

radioactive adj.
放射性的
radioactive fallout across Europe

fallout n.
放射性尘降物
radioactive fallout across Europe

inhabitant n.
居民
about 45,000 inhabitants

comparable adj.
相当的；可比较的
comparable in density

density n.
密度
comparable in density

contamination n.
污染
free of contamination

fieldwork n.
实地研究
Later fieldwork bolstered

bolster v.
支持；加强证据
fieldwork bolstered these findings

hazardous adj.
危险的
the most hazardous spots

absence n.
不存在；缺少
the absence of agriculture

organism n.
生物
Smaller organisms fared worse

exposure n.
辐射暴露
where exposure was greater

decompose v.
分解；腐烂
leaves decomposing about 40 per cent

impede v.
阻碍
impeding the return of nutrients

guarantee v.
保证
guarantee the health

continual adj.
持续的
low but continual doses

inflict v.
造成伤害
low but continual doses inflict

contentious adj.
有争议的
remains contentious

dividend n.
好处；红利
the dividend of human departure

sluggish adj.
缓慢的
the sluggish rot of fallen leaves

文中词汇与搭配

补充词汇共43项。保留每个表达在本文中的意思。

disaster n.
灾难

startling adj.
惊人的

wildlife n.
野生动物

reactor n.
核反应堆

be emptied of 动词短语
被撤空；人员被全部迁走

radius n.
半径

off-limits adj.
禁止进入的

fence off 动词短语
围封；用围栏隔开

adjoining adj.
相邻的

terrain n.
地带；地形

Current Biology 专名
《当代生物学》

aerial adj.
空中的

footprint n.
足迹

elk n.
驼鹿

roe deer 名词短语
狍

boar n.
野猪

to prep.
与；引出比较对象

reserve n.
自然保护区

numerous adj.
数量众多的

helicopter n.
直升机

elk n.
驼鹿

surge v.
迅速增加

文中词汇与搭配 续

补充词汇共43项。保留每个表达在本文中的意思。

post-disaster adj.
灾后的

snap v.
拍下照片

suppress v.
抑制；压低

probable adj.
很可能的

mundane adj.
普通的；平常的

agriculture n.
农业

forestry n.
林业

nest v.
筑巢

intense adj.
强烈的

pollinator n.
传粉动物

plentiful adj.
数量多的；丰富的

polluted adj.
受污染的

individual adj.
个体的

how much damage 名词短语
多大伤害

clash over 动词短语
在……问题上存在分歧

burgeoning adj.
迅速增长的

herd n.
兽群

trumpet v.
大力宣扬；彰显

departure n.
离开

rot n.
腐烂

ledger n.
账本；此处比喻记录

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Chernobyl is _____ disaster, yet it holds a startling amount of wildlife.

02 Later fieldwork _____ these findings.

03 The probable reason is mundane: the _____ of agriculture, forestry, and hunting.

04 Counting heads cannot _____ the health of individual animals

文章主线

- 灾后 1986年切尔诺贝利核事故后，居民撤离、周边封禁，受污染地区却出现大量野生动物。
- 繁盛 白俄罗斯一侧多种兽类密度与四个未污染保护区相当，狼的数量为对照区的七倍多。
- 原因 可能的解释是农耕、伐木和狩猎消失，人类活动减少给野生动物带来了生存空间。
- 代价 部分研究发现高辐射处筑巢鸟数量及鸟种较少，传粉昆虫和蜘蛛也少；最重污染处落叶少分解约40%，阻碍养分回土。
- 限度 动物多不等于个体健康；长期低剂量辐射造成多大伤害仍有争议，不能据此说辐射无害。

参考答案　1 synonymous with 2 bolstered 3 absence 4 guarantee

十一天不睡觉的少年

Eleven Days Without Sleep



十一天不睡觉，身体会怎样？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On 28 December 1963 Randy Gardner, a 17-year-old science enthusiast at a San Diego high school, set out to forgo slumber for a regional science fair. Two companions diligently jolted him awake whenever his alertness waned and logged his mood. The project, a voluntary endeavour, went on to take first place.

William Dement, a sleep researcher at Stanford, read press coverage of the attempt and was intrigued. He drove to San Diego and assisted John J. Ross, a Navy researcher, in monitoring the sleepless boy closely. The two investigators attentively noted the fluctuations in his state as the deprivation progressively deepened.

The consequences soon mounted. Gardner grew moody, lost concentration, and found his short-term memory impaired. He became suspicious, at times paranoid, and began to hallucinate. On the fourth day he was gripped by the delusion that he was a professional American football player.

Gardner ended the wearying attempt on 8 January 1964, after 264.4 hours, about 11 days and 25 minutes. He then slept soundly for about 14 hours and woke apparently refreshed. Researchers probed the case in a 1965 scholarly paper, which helped highlight microsleeps, involuntary lapses into sleep lasting a few seconds. Gardner may have had such momentary bouts without noticing.

In 1997 Guinness World Records discontinued monitoring such attempts because of the health perils, and others have since alleged longer stints awake. In a 2017 interview Gardner said severe insomnia had tormented him for about a decade, and he himself pinned the blame on the experiment. The boy who had seemed to bounce back from exhaustion with one long rest would, decades later, barely doze off.

中文译文

1963年12月28日，圣迭戈一所高中里，17岁的科学爱好者兰迪·加德纳为参加地区科学展，开始尝试不睡觉。每当他的清醒程度下降，两名同伴就尽力把他唤醒，并记录他的情绪。这项自愿开展的尝试后来获得了第一名。

斯坦福大学睡眠研究者威廉·德门特读到有关报道后产生了兴趣。他开车来到圣迭戈，协助海军研究员约翰·J.罗斯密切观察这名不眠的少年。随着缺觉程度逐步加重，两位研究者仔细记录着他状态的变化。

不良后果很快加剧。加德纳变得喜怒无常，注意力难以集中，短时记忆也受到损害。他开始多疑，有时甚至出现偏执妄想，还产生了幻觉。到第四天，他竟认定自己是一名职业美式橄榄球运动员。

1964年1月8日，加德纳结束了这场令人疲惫的挑战，历时264.4小时，约合11天零25分钟。之后，他酣睡约14小时，醒来时似乎恢复了精神。研究者在1965年的一篇学术论文中探讨了这个案例，使人们更加关注“微睡眠”：人会不由自主地短暂入睡，持续几秒钟。加德纳可能经历过这种短暂睡眠，自己却没有察觉。

1997年，吉尼斯世界纪录因健康风险停止监测此类挑战，此后仍有人声称自己保持清醒的时间更长。2017年接受采访时，加德纳说，严重失眠已折磨了自己约十年，他本人将其归咎于当年的实验。那个曾经看似睡个长觉便能从极度疲惫中恢复的少年，几十年后却几乎连打个盹都做不到。

晚年失眠与当年实验的因果关系，文中并未证实。

核心词汇

先记这22项。连着下方的原文短语一起回想。

enthusiast n.
爱好者
a 17-year-old science enthusiast

forgo v.
放弃；不享用
to forgo slumber

alertness n.
警觉；清醒程度
his alertness waned

wane v.
减弱；衰退
his alertness waned

voluntary adj.
自愿的
a voluntary endeavour

endeavour n.
尝试；努力
a voluntary endeavour

intrigue v.
使感兴趣
was intrigued

fluctuation n.
波动；变化
the fluctuations in his state

deprivation n.
剥夺；缺失
the deprivation progressively deepened

concentration n.
注意力；专注
lost concentration

impair v.
损害；使受损
short-term memory impaired

paranoid adj.
偏执多疑的；有妄想的
at times paranoid

hallucinate v.
产生幻觉
began to hallucinate

delusion n.
妄想；错误的确信
gripped by the delusion

involuntary adj.
不由自主的
involuntary lapses into sleep

discontinue v.
停止；终止
discontinued monitoring such attempts

peril n.
危险；风险
because of the health perils

insomnia n.
失眠
severe insomnia

torment v.
折磨；使痛苦
had tormented him for about a decade

pin the blame on 动词短语
把责任归咎于
pinned the blame on the experiment

bounce back from 动词短语
从……中迅速恢复
bounce back from exhaustion

exhaustion n.
精疲力竭
bounce back from exhaustion

文中词汇与搭配

补充词汇共36项。保留每个表达在本文中的意思。

slumber n.
睡眠

regional adj.
地区的；区域性的

companion n.
同伴

diligently adv.
尽心尽力地

jolt v.
使猛然惊醒

awake adj.
醒着的

mood n.
情绪

take first place 动词短语
获得第一名

coverage n.
新闻报道

assist v.
协助；帮助

in prep.
在……方面（assist…in）

sleepless adj.
不眠的；未睡的

closely adv.
密切地

investigator n.
研究者；调查人员

attentively adv.
专注地；仔细地

progressively adv.
逐步地；不断地

deepen v.
加深；加重

consequence n.
后果

文中词汇与搭配 续

补充词汇共36项。保留每个表达在本文中的意思。

mount v.
增加；加剧

moody adj.
喜怒无常的

short-term adj.
短期的；短时的

suspicious adj.
多疑的；猜疑的

be gripped by 动词短语
被强烈情绪或想法控制

wearying adj.
令人疲惫的

soundly adv.
（睡得）熟；酣畅地

refreshed adj.
恢复精神的

probe v.
深入探究

scholarly adj.
学术的

highlight v.
使受到关注；凸显

lapse into 动词短语
不由自主地陷入

momentary adj.
短暂的；瞬间的

bout n.
一阵；一段发作

allege v.
声称（尚未证实）

stint n.
一段时间

barely adv.
几乎不；勉强

doze off 动词短语
打瞌睡；短暂入睡

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Two companions diligently jolted him awake whenever his _____ waned and logged his mood.

02 The project, a voluntary _____, went on to take first place.

03 Gardner grew moody, lost concentration, and found his short-term memory _____.

04 He became suspicious, at times paranoid, and began to _____.

文章主线

- 挑战 1963年底，17岁加德纳为科学展尝试不睡觉；同伴帮他保持清醒，睡眠研究者随后加入监测。
- 损害 他逐渐情绪不稳，注意力和短时记忆受损，出现多疑和幻觉，甚至以为自己是职业橄榄球员。
- 结束 264.4小时（约11天）后挑战结束；酣睡约14小时后，他醒来似乎恢复了精神。
- 疑点 他可能经历过未察觉的数秒微睡眠；吉尼斯1997年因健康风险停止记录此类挑战。
- 晚年 2017年，他说严重失眠已折磨自己约十年，并归咎于当年实验；文章没有证明这种因果关系。

参考答案　1 alertness 2 endeavour 3 impaired 4 hallucinate

集装箱：重塑世界的铁盒子

The Steel Box That Remade the World



一个普通铁箱，怎样改变了全球货运？

英文原文

On 26 April 1956 a converted Second World War tanker, the Ideal X, slipped out of Port Newark, New Jersey, for Houston. Clamped to her deck were 58 containers, trailer bodies detached from their wheels and chassis. Few spectators would have taken so unremarkable a cargo for a landmark.

At the time, freight travelled loose. Gangs of twenty or more dockworkers carried sacks, barrels, and crates piece by piece. It was strenuous drudgery. Ships could spend up to half their time berthed, and theft was rampant. In 1956 loading loose cargo cost about $5.83 a ton; loading the Ideal X cost, by the company's reckoning, about 16 cents.

Containers were not new. The hurdle was consensus: rival firms' boxes had incompatible dimensions and latches. An international standards committee formed in 1961. In January 1963 Sea-Land, the Ideal X's renamed operator, let the committee use its patents free of royalties, including those on corner fittings for hoisting. Lifted by crane, a box could be transferred smoothly between ship, lorry, and train.

This efficiency came at the expense of livelihoods. New York's waterfront employment shrank by about three quarters between 1963 and 1976. Much of the cargo crossed the harbour to a new terminal at Elizabeth, inaugurated in August 1962. Most of London's historic docks closed one after another from 1967 to 1980, while Felixstowe, on the Suffolk coast, expanded.

The rewards were substantial. A study published in the Journal of International Economics in 2016 estimated that containerisation had lifted trade between industrialised countries by a cumulative 700 per cent over 20 years. Tireless cranes now lift identical containers into orderly, immense stacks.

中文译文

1956年4月26日，一艘由二战油轮改装而成的“理想X号”悄然驶出新泽西州纽瓦克港，前往休斯敦。甲板上固定着58个集装箱：它们原是挂车车厢，已从车轮和底盘上拆下。几乎没有旁观者会想到，这批平淡无奇的货物将成为一个里程碑。

当时，货物都是散装运输。每组二十人或更多的码头工人，一件件搬运麻袋、木桶和板条箱。这是繁重而枯燥的苦差事。船舶多达一半的时间可能耗在泊港上，偷窃也十分猖獗。1956年，装载散货每吨约需5.83美元；而据运营公司估算，“理想X号”的装船费用每吨仅约16美分。

集装箱并非新事物，障碍在于达成共识：竞争公司的箱子尺寸与锁扣互不兼容。1961年，一个国际标准委员会成立。1963年1月，“理想X号”的运营方——当时已更名为海陆公司——允许委员会免付专利使用费，使用其专利，其中包括用于吊装的角配件专利。吊机把箱子吊起后，便能在船舶、卡车和火车之间顺畅转运。

这种效率提升也以人们的生计为代价。1963至1976年，纽约滨水区的就业人数缩减了约四分之三。大量货运转移到港湾对面的伊丽莎白新码头；该码头于1962年8月启用。1967至1980年，伦敦大多数历史悠久的码头相继关闭，萨福克海岸的费利克斯托港却不断扩张。

回报同样可观。2016年发表于《国际经济学杂志》的一项研究估计，集装箱化在20年间使工业化国家间的贸易累计增长了约700%。如今，不知疲倦的吊机把规格一致的集装箱吊起，码成整齐而庞大的箱垛。

核心词汇

先记这22项。连着下方的原文短语一起回想。

cargo n.
船运货物
so unremarkable a cargo

landmark n.
里程碑；重大转折
a cargo for a landmark

freight n.
货物；货运
freight travelled loose

drudgery n.
单调繁重的苦差事
strenuous drudgery

rampant adj.
猖獗的；泛滥的
theft was rampant

reckoning n.
估算
by the company's reckoning

hurdle n.
障碍；难关
The hurdle was consensus

consensus n.
共识
The hurdle was consensus

incompatible adj.
互不兼容的
incompatible dimensions and latches

dimension n.
尺寸
incompatible dimensions and latches

patent n.
专利
use its patents free of royalties

royalty n.
专利使用费
use its patents free of royalties

crane n.
起重机；吊机
Lifted by crane

transfer v.
转运；转移
could be transferred smoothly

efficiency n.
效率
This efficiency came

at the expense of 介词短语
以……为代价
at the expense of livelihoods

livelihood n.
生计
at the expense of livelihoods

employment n.
就业；就业人数
waterfront employment shrank

inaugurate v.
正式启用
inaugurated in August 1962

substantial adj.
可观的；大量的
The rewards were substantial

cumulative adj.
累计的
a cumulative 700 per cent

identical adj.
完全相同的
lift identical containers

文中词汇与搭配

补充词汇共43项。保留每个表达在本文中的意思。

convert v.
改装；改造

tanker n.
油轮

slip out of 动词短语
悄然驶离

clamp v.
夹紧；固定

container n.
集装箱

trailer n.
挂车

detach from 动词短语
从……拆下

wheel n.
车轮

chassis n.
车辆底盘

spectator n.
旁观者

unremarkable adj.
平淡无奇的

loose adj.
散装的；未捆装的

gang n.
一组工人

dockworker n.
码头工人

sack n.
大袋；麻袋

barrel n.
大桶

crate n.
板条箱

strenuous adj.
繁重费力的

up to 介词短语
多达

berth v.
停泊；靠泊

theft n.
偷窃

load v.
装载

文中词汇与搭配 续

补充词汇共43项。保留每个表达在本文中的意思。

by prep.
据；按照

rival adj.
竞争对手的

latch n.
锁扣；门闩

international standard 名词短语
国际标准

fitting n.
配件

hoist v.
吊起；吊装

smoothly adv.
顺畅地

lorry n.
卡车

waterfront n.
滨水区

shrink v.
缩减

terminal n.
货运码头

historic adj.
历史悠久的

dock n.
码头

expand v.
扩张

reward n.
回报

economics n.
经济学

industrialised adj.
工业化的

tireless adj.
不知疲倦的

orderly adj.
整齐有序的

immense adj.
巨大的

stack n.
垛；整齐堆叠的一摞

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 The hurdle was _____:

02 and theft was _____.

03 rival firms' boxes had _____ dimensions and latches.

04 This efficiency came at the expense of _____.

文章主线

- 启航 过去码头逐件搬散货，船常久停；1956年，理想X号载58个集装箱启航。
- 降本 据运营方估算，该船每吨装船费约0.16美元，远低于当时散货的5.83美元。
- 互通 各家箱型原不兼容；运营方海陆公司免费开放专利，推动标准统一，箱子可在船、卡车和火车间转运。
- 转移 货运转向新港，老码头和工人生计受冲击；1963至1976年，纽约滨水区就业缩减约四分之三。
- 回报 2016年一项研究估计，集装箱化在20年间使工业化国家间贸易累计增加约700%。

参考答案　1 consensus 2 rampant 3 incompatible 4 livelihoods

永不沉没的神话

The Myth of the Unsinkable



守法的泰坦尼克号，为何仍救不了所有人？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On 10 April 1912 the opulent liner Titanic left Southampton on her maiden voyage. A 1911 trade journal had dubbed her and her sister ship "practically unsinkable". At about 11.40pm on 14 April she collided with an iceberg; by 2.20am she had vanished beneath the waves, and the death toll would surpass 1,500.

Folklore blames arrogance. Yet the Titanic was compliant with the law. Under 1894 rules, Britain's maritime regulator pegged lifeboat numbers to gross tonnage (volume, not weight) on a scale that topped out at 16 boats for 10,000 tons or more. At about 46,000 tons she carried 20, exceeding the requirement. Their capacity was 1,178, roughly half the 2,200 aboard: sufficient in law, inadequate at sea.

Within radio range, the steamer Californian kept an equally lawful routine. Her sole wireless operator, Cyril Evans, was on the payroll of the Marconi company; when he slept, so did the radio. Around 11.30pm he switched off his set. Hindsight may call that negligence, yet no rule obliged him to stay vigilant, so the Titanic's frantic distress calls went unheard there.

Between April and July 1912, inquiries in Washington and London heard testimony from survivors and the Californian's crew. That July, American legislation made round-the-clock radio watch compulsory. In January 1914 delegates of 13 nations signed the first Safety of Life at Sea convention, stipulating lifeboat seats for all, continuous radio watch, and an ice patrol. The First World War stalled the treaty; the patrol went ahead regardless.

The doomed liner sank within three hours. The outdated rules she sailed under had stayed afloat since 1894; it took her dead to discredit them.

中文译文

1912年4月10日，豪华邮轮泰坦尼克号驶离南安普敦，开始首航。1911年，一本行业期刊曾将它和姊妹船称为“几乎不会沉没的船”。4月14日晚约11时40分，它撞上一座冰山；到次日凌晨2时20分，船已沉入海中，最终遇难人数超过1500。

坊间说法将灾难归咎于傲慢。然而，泰坦尼克号符合当时的法律。根据1894年的规定，英国海事监管机构按总吨位确定救生艇数量；总吨位衡量的是容积，并非重量。这套分级标准的上限，是对1万总吨及以上的船只要求配备16艘救生艇。泰坦尼克号约4.6万总吨，带了20艘，超过法定要求。但这些艇只能容纳1178人，约为船上2200人的一半：法律上足够，海上救生却不足。

在无线电通信范围内，加州人号轮船的值班惯例同样没有违规。船上唯一的报务员西里尔·埃文斯受雇于马可尼公司；他一休息，电台也随之无人值守。晚上约11时30分，他关掉了设备。事后看来，这或许可称为疏忽，但当时没有规定要求他保持值守，因此泰坦尼克号急切的求救信号在加州人号上无人听到。

1912年4月至7月，华盛顿和伦敦的调查听取了幸存者和加州人号船员的证词。同年7月，美国立法强制要求全天候无线电值守。1914年1月，13个国家的代表签署首部《海上人命安全公约》，规定救生艇座位必须足够全员使用、电台连续值守，并开展冰情巡逻。第一次世界大战使公约搁置，冰情巡逻却仍然开展起来。

这艘在劫难逃的邮轮不到三小时就沉没了。它所遵循的过时规定，却自1894年起一直照常施行；直到船上众多生命逝去，这些规定才失去了公信力。

总吨位衡量容积；“没有违规”并不表示法律要求报务员关机。

核心词汇

先记这22项。连着下方的原文短语一起回想。

opulent adj.
豪华的；奢华的
the opulent liner Titanic

maiden voyage 名词短语
首航
on her maiden voyage

dub v.
把……称为
had dubbed her and her sister ship

collide with 动词短语
与……相撞
she collided with an iceberg

death toll 名词短语
遇难人数
the death toll would surpass 1,500

arrogance n.
傲慢；自大
Folklore blames arrogance

compliant with 形容词短语
符合；遵守
was compliant with the law

regulator n.
监管机构
Britain's maritime regulator

gross tonnage 名词短语
总吨位（衡量船内容积）
gross tonnage (volume, not weight)

capacity n.
容纳人数；容量
Their capacity was 1,178

sufficient adj.
足够的
sufficient in law

inadequate adj.
不足的；不够的
inadequate at sea

hindsight n.
事后认识；后见之明
Hindsight may call that negligence

negligence n.
疏忽；失职
Hindsight may call that negligence

oblige v.
强制；要求
no rule obliged him to stay vigilant

vigilant adj.
警觉的；保持值守的
no rule obliged him to stay vigilant

testimony n.
证词；证言
heard testimony from survivors

legislation n.
立法；法律
American legislation

compulsory adj.
强制的；必须履行的
radio watch compulsory

convention n.
国际公约
Safety of Life at Sea convention

stipulate v.
规定；明确要求
stipulating lifeboat seats for all

discredit v.
使失去信誉；证明不可靠
to discredit them

文中词汇与搭配

补充词汇共37项。保留每个表达在本文中的意思。

liner n.
大型客轮；邮轮

trade journal 名词短语
行业期刊

practically adv.
几乎；实际上

unsinkable adj.
不会沉没的

iceberg n.
冰山

vanish v.
消失

surpass v.
超过

folklore n.
坊间流传的说法

maritime adj.
海事的；海上的

peg v.
使挂钩；按某标准确定

to prep.
与……（peg…to搭配）

scale n.
分级标准；等级表

top out at 动词短语
最高达到；上限为

exceed v.
超过

requirement n.
规定的要求

roughly adv.
大约；大致

equally adv.
同样地

lawful adj.
合法的；未违规的

文中词汇与搭配 续

补充词汇共37项。保留每个表达在本文中的意思。

sole adj.
唯一的

wireless adj.
无线电的

operator n.
操作员；报务员

on the payroll of 介词短语
受雇于；领取……的薪酬

switch off 动词短语
关掉（设备）

frantic adj.
慌乱而急切的

distress call 名词短语
求救呼叫；求救信号

unheard adj.
未被听见的

survivor n.
幸存者

round-the-clock adj.
全天候的；不间断的

delegate n.
代表

continuous adj.
连续的；不间断的

patrol n.
巡逻

stall v.
使暂缓；使停滞

treaty n.
条约

regardless adv.
尽管如此；仍然

doomed adj.
在劫难逃的

outdated adj.
过时的

stay afloat 动词短语
维持有效；继续存在（比喻）

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Yet the Titanic was _____ the law.

02 Their capacity was 1,178, roughly half the 2,200 aboard: sufficient in law, _____ at sea.

03 Hindsight may call that _____, yet no rule obliged him to stay vigilant, so the Titanic's frantic distress calls went unheard there.

04 That July, American legislation made round-the-clock radio watch _____.

文章主线

- 沉船 1912年，号称几乎不沉的泰坦尼克号首航撞冰山，不足三小时沉没，逾1500人遇难。
- 救生 旧规按总吨位（容积）定艇数，万吨以上只需16艇；它有20艇、1178座，仅够船上2200人约一半。
- 守听 附近加州人号的唯一报务员关机休息并未违规，电台没听到求救；当时没有全天值守的强制规定。
- 改革 1912年美国强制全天守听；1914年13国签约，要求全员救生艇座位、连续守听和冰情巡逻。
- 教训 一战使公约搁置，冰情巡逻仍启动；沿用自1894年的旧规暴露出：合法配置，并不等于安全足够。

参考答案　1 compliant with 2 inadequate 3 negligence 4 compulsory

法老的诅咒

The Curse of the Pharaoh



法老的诅咒，经得起证据检验吗？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On 4 November 1922, in the Valley of the Kings, labourers engaged by the archaeologist Howard Carter uncovered a staircase carved into the rock. On 26 November, with his wealthy sponsor Lord Carnarvon beside him, Carter made a small hole in the sealed doorway and glimpsed the glittering gold of the boy king Tutankhamun. The spectacular find riveted the world.

Carnarvon sold exclusive rights to The Times, a monopoly that left competing reporters starved of news. Many resorted to speculation or sought outside commentators. In March 1923 the best-selling novelist Marie Corelli warned publicly that dire punishment awaited anyone who entered a sealed tomb. Within a fortnight, her words would seem uncanny.

That month a mosquito bit Carnarvon. He nicked the bite with a razor while shaving, it turned septic, and blood poisoning led to pneumonia. He died in Cairo on 5 April 1923. Newspapers reported that the city's lights failed at that instant and that his dog in England howled and died. Yet blackouts were almost a daily occurrence in Cairo.

No curse inscription was ever found in the tomb. In December 2002 a British Medical Journal study traced the fate of 44 Westerners in Egypt at the time. The 25 exposed to the tomb died at an average age of 70, versus 75 for the rest, a margin not statistically significant. The mortality figures lent the ominous rumour no credibility.

Carnarvon's untimely demise had fuelled the tale, giving it impetus. Carter, the excavator who had unsealed the burial place, died in March 1939, aged 64, almost sixteen years after his patron. The legend had left him conspicuously unharmed.

中文译文

1922年11月4日，在埃及帝王谷，考古学家霍华德·卡特雇用的工人发现了一段凿入岩石的阶梯。11月26日，富有的赞助人卡纳冯勋爵站在身旁，卡特在封住的入口上开了一个小孔，瞥见少年国王图坦卡蒙墓中闪耀的黄金。这一惊人发现吸引了全世界的目光。

卡纳冯将独家报道权卖给《泰晤士报》，使其他竞争记者严重缺少消息。许多人转而猜测，或向外界评论者寻求说法。1923年3月，畅销小说家玛丽·科雷利公开警告，任何进入封闭陵墓的人都会受到可怕的惩罚。不到两周，她的话就将显得诡异。

当月，一只蚊子叮了卡纳冯。他刮胡子时用剃刀划破叮咬处，伤口感染化脓，随后败血症引发肺炎。1923年4月5日，他在开罗去世。报纸声称，就在他去世的一刻，开罗全城停电，他在英国的狗也嚎叫着死去。然而，停电在当时的开罗几乎天天发生。

墓中从未发现过诅咒铭文。2002年12月，《英国医学杂志》的一项研究追溯了当时身在埃及的44名西方人的命运。接触过陵墓的25人平均享年70岁，其余人平均75岁，这一差距在统计上并不显著。死亡统计并未为那则不祥的传闻提供可信依据。

卡纳冯的过早离世助长了诅咒传说，使它越传越广。但亲自打开陵墓的发掘者卡特，直到1939年3月才去世，享年64岁，比赞助人的去世时间晚了近16年。这位开墓者却明显没有遭遇传说中的灾祸。

近16年指两人去世日期的间隔，并不是两人的寿命相差16年。

核心词汇

先记这22项。连着下方的原文短语一起回想。

sponsor n.
赞助人；资助者
his wealthy sponsor Lord Carnarvon

rivet v.
牢牢吸引注意力
The spectacular find riveted the world

exclusive adj.
独家的；专属的
sold exclusive rights

monopoly n.
垄断；独占
a monopoly

be starved of 动词短语
严重缺少
reporters starved of news

resort to 动词短语
转而采用；诉诸
resorted to speculation

speculation n.
猜测；推测
resorted to speculation

dire adj.
可怕的；极其严重的
dire punishment

fortnight n.
两周
Within a fortnight

uncanny adj.
诡异的；不可思议的
her words would seem uncanny

turn septic 动词短语
感染化脓
it turned septic

pneumonia n.
肺炎
blood poisoning led to pneumonia

inscription n.
铭文；刻文
No curse inscription

be exposed to 动词短语
接触到
The 25 exposed to the tomb

margin n.
差距；差额
a margin not statistically significant

significant adj.
（统计上）显著的
not statistically significant

mortality n.
死亡情况；死亡统计
The mortality figures

credibility n.
可信度
no credibility

demise n.
去世；死亡
Carnarvon's untimely demise

fuel v.
助长；促使加剧
had fuelled the tale

impetus n.
推动力
giving it impetus

conspicuously adv.
显眼地；格外醒目地
conspicuously unharmed

文中词汇与搭配

补充词汇共32项。保留每个表达在本文中的意思。

labourer n.
工人；体力劳动者

engage v.
雇用；聘请

uncover v.
发现；发掘

staircase n.
楼梯；阶梯

carve into 动词短语
雕凿于；刻入

wealthy adj.
富有的

doorway n.
门口；入口

glittering adj.
闪闪发光的

spectacular adj.
惊人的；壮观的

competing adj.
相互竞争的

commentator n.
评论者

best-selling adj.
畅销的

punishment n.
惩罚

tomb n.
陵墓

mosquito n.
蚊子

nick v.
轻轻划破；割破

razor n.
剃刀

shave v.
刮胡子

at that instant 介词短语
就在那一瞬间

howl v.
嚎叫

blackout n.
停电

occurrence n.
发生的事；事件

fate n.
命运；遭遇

versus prep.
与……相比

ominous adj.
不祥的

rumour n.
传闻；谣言

untimely adj.
过早的；不合时宜的

excavator n.
发掘者

unseal v.
启封；打开封口

burial n.
埋葬；墓葬

patron n.
资助人；赞助人

unharmed adj.
未受伤害的

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Carnarvon sold exclusive rights to The Times, a _____ that left competing reporters starved of news.

02 Many resorted to _____ or sought outside commentators.

03 No curse _____ was ever found in the tomb.

04 The mortality figures lent the ominous rumour no _____.

文章主线

- 发现 1922年，卡特与赞助人卡纳冯发现图坦卡蒙墓，墓中珍宝震惊世界。
- 传言 《泰晤士报》买断报道，其他记者转向猜测；卡纳冯次年去世，诅咒故事随之盛行。
- 病因 卡纳冯剃破蚊叮处，感染后发生败血症和肺炎；报纸所述停电，在开罗本就几乎天天发生。
- 对照 研究追踪44人：接触墓的25人平均活70岁，其余75岁，这点差距仍可能来自偶然。
- 结论 墓中从未发现诅咒铭文；卡纳冯死后，亲自开墓的卡特又活了近16年，现有证据不支持诅咒。

参考答案　1 monopoly 2 speculation 3 inscription 4 credibility

孩子们发明的语言

The Language the Children Invented



没有共同手语的孩子，怎样创造出一门语言？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On a rattling bus in Managua, deaf pupils kept up a brisk, voiceless hubbub across the aisle, palms fluttering, wrists flicking. Their teachers could not follow a word.

Before the late 1970s Nicaragua had no deaf community. Most deaf people led secluded lives, sharing a homemade repertoire of gestures with parents and siblings. In 1977 a special-education centre opened in Managua's San Judas district with 50 deaf pupils. By 1983 the centre and a vocational school for teenage pupils held more than 400.

The curriculum hinged on oral Spanish and lipreading, and teachers signed only the alphabet. Pupils studied lips in vain, and most failed to grasp what a Spanish word was. At recess, in the street, and on the buses, they mingled gestures with home signs. A rough jumble ripened rapidly into a coherent language entirely their own.

The teachers found it unintelligible, and in June 1986 the Ministry of Education brought in Judy Kegl, an American Sign Language linguist from MIT. A 2001 study found that pupils who arrived younger, and later intakes, used grammatical devices more often and more uniformly than their forerunners. Each fresh wave of young signers added complexity.

In a 2004 study, gesturing Spanish speakers and the earliest signers described something rolling downhill by welding how it moved and where it went into one undivided movement. Later signers dismantled it into discrete segments in linear order, a framework found nowhere in the gestures around them.

Linguists had a rare chance to watch a language grow from infancy to maturity. Its builders were not teachers or experts but deaf playmates in playgrounds, streets, and buses.

中文译文

马那瓜一辆颠簸作响的公车上，聋人学生隔着过道热烈地“聊”着天，手掌挥动，手腕轻摆，热闹却无声。老师们一个字也看不懂。

20世纪70年代末以前，尼加拉瓜尚未形成聋人社群。多数聋人生活闭塞，只与父母、兄弟姐妹共用一套自创手势。1977年，一所特殊教育中心在马那瓜圣胡达斯区开办，招收了50名聋人学生。到1983年，这所中心与一所面向青少年学生的职业学校，共有400多名学生。

课程以西班牙语口语和唇读为中心，老师只用手势比划字母。学生努力观察说话时的嘴唇，却徒劳无功；多数人连一个西班牙语单词是什么都没能弄明白。但在课间、街头和公车上，他们把一般手势与各自家中的手势混合起来。一套粗糙杂乱的表达迅速发展为一门连贯、自成体系的语言。

老师们看不懂这种语言。1986年6月，教育部请来麻省理工学院研究美国手语的语言学家朱迪·凯格尔。2001年的一项研究发现，与较早的学生相比，入学时年龄更小的学生，以及后来几批入学者，使用语法手段更频繁，也更统一。每一批新的年幼手语使用者，都为这门语言增添了复杂性。

2004年的一项研究发现，打手势的西班牙语使用者和最早一批手语使用者，在描述某个东西滚下山坡时，会把“怎样运动”和“往哪里去”合成一个不可分的动作。后来的手语使用者则把它拆成彼此独立的片段，按先后顺序表达；这种结构在他们周围人的手势中无处可寻。

语言学家难得有机会亲眼看着一门语言从初创走向成熟。创造它的不是教师或专家，而是在操场、街头和公车上一起玩耍、交流的聋人孩子。

核心词汇

先记这22项。连着下方的原文短语一起回想。

community n.
社群
no deaf community

secluded adj.
与外界隔绝的
led secluded lives

repertoire n.
掌握的一整套表达手段
a homemade repertoire of gestures

curriculum n.
课程；课程设置
The curriculum hinged on oral Spanish

hinge on 动词短语
以……为中心；取决于
hinged on oral Spanish and lipreading

in vain 介词短语
徒劳地
Pupils studied lips in vain

grasp v.
理解；领会
failed to grasp

mingle v.
混合；融合
mingled gestures with home signs

coherent adj.
连贯的；自成体系的
a coherent language

unintelligible adj.
无法理解的
The teachers found it unintelligible

linguist n.
语言学家
an American Sign Language linguist

intake n.
新招收的几批学生
later intakes

grammatical adj.
语法的
used grammatical devices

uniformly adv.
一致地；统一地
more uniformly than their forerunners

complexity n.
复杂性
added complexity

dismantle v.
拆解
dismantled it into discrete segments

discrete adj.
分立的；彼此可区分的
discrete segments in linear order

segment n.
片段
discrete segments in linear order

linear order 名词短语
线性顺序；依次排列
segments in linear order

framework n.
结构；框架
a framework found nowhere

infancy n.
初创阶段；幼年期
from infancy to maturity

maturity n.
成熟阶段
from infancy to maturity

文中词汇与搭配

补充词汇共41项。保留每个表达在本文中的意思。

rattle v.（分词作定语）
颠簸作响

brisk adj.
轻快热烈的

voiceless adj.
无声的

hubbub n.
喧闹；热闹的交流

aisle n.
过道

palm n.
手掌

flutter v.
挥动；轻快拍动

wrist n.
手腕

flick v.
轻快地摆动

homemade adj.
自创的；自家制作的

sibling n.
兄弟姐妹

special education n.（作定语）
特殊教育

district n.
地区；区

pupil n.
学生

vocational school 名词短语
职业学校

teenage adj.
青少年的

oral adj.
口语的

lipreading n.
唇读；看口形辨话

alphabet n.
字母表

lip n.
嘴唇；此处涉及说话口形

文中词汇与搭配 续

补充词汇共41项。保留每个表达在本文中的意思。

at recess 介词短语
在课间休息时

with prep.
与；同

rough adj.
粗糙的；尚未完善的

jumble n.
杂乱的混合物

ripen v.
逐渐发展成熟

rapidly adv.
迅速地

into prep.
成为；用于引出变化结果

entirely adv.
完全地

Ministry of Education 专名
教育部

bring in 动词短语
请来；引入

American Sign Language 专名
美国手语

MIT 专名
麻省理工学院

forerunner n.
较早的一批人；前辈

signer n.
手语使用者

roll v.
滚动

downhill adv.
向坡下

weld v.
熔合；把两部分合为一体

undivided adj.
未分开的；不可分的

nowhere adv.
无处

builder n.
创造者

playmate n.
玩伴

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 A rough jumble ripened rapidly into a _____ language entirely their own.

02 The curriculum _____ oral Spanish and lipreading, and teachers signed only the alphabet.

03 Each fresh wave of young signers added _____.

04 Later signers dismantled it into discrete segments in _____, a framework found nowhere in the gestures around them.

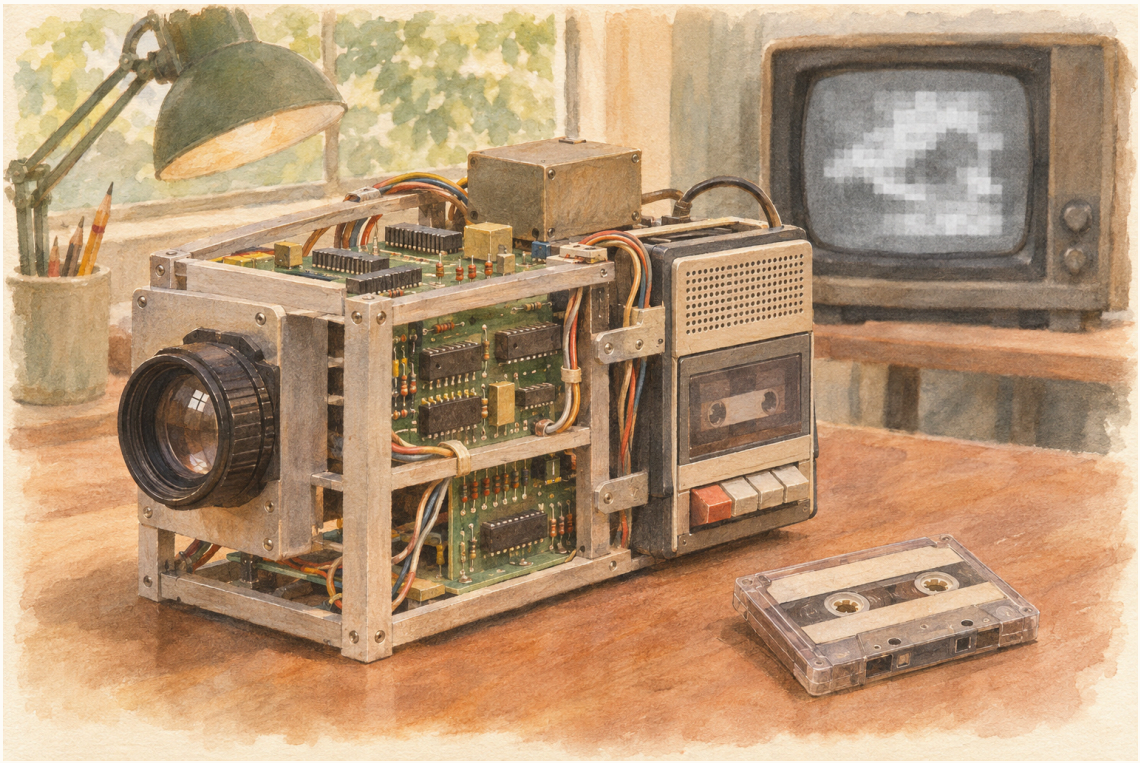
文章主线

- 相聚 1977年，尼加拉瓜学校把聋童聚在一起；此前他们多只与家人用自创手势交流。
- 受阻 学校教西班牙语口语和唇读，多数学生未能掌握；课外，孩子们开始用手势相互交流。
- 自创 孩子在课间、街头和公车上交流，把自家手势混合起来，逐渐形成了一门共同语言。
- 演变 入学年龄更小的孩子和后来几批新生使语法更统一复杂；后来的孩子还把同一手势里的怎么动、往哪去拆开，按先后顺序表达。
- 意义 这门语言出自孩子的同伴交流，而非专家传授；语言学家得以观察它逐渐成熟。

参考答案　1 coherent 2 hinged on 3 complexity 4 linear order

发明了数码相机，却被它打败

The Company That Invented Its Own Killer



造出数码相机的公司，为什么仍被数码时代冲击？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In 1975 Kodak engineer Steven Sasson built the first digital camera. The toaster-sized gadget, a cumbersome heavyweight of about eight pounds, was hardly portable, let alone handheld. Its first rudimentary picture held 10,000 black-and-white pixels. Recording it on a cassette took 23 seconds, and a playback unit showed it on a television.

Kodak's revenues showed its dependence on film, the profitable pillar of its business. Technical staffers were appreciative of the camera, but it needed no film. By Sasson's recollection, his superiors at headquarters called it cute and urged him to keep it hushed. A patent followed in 1978. Production never did.

The advent of digital photography proved disruptive. In June 2009 Kodak said it would stop making Kodachrome, its analogue film, citing lessened demand in the digital era. Kodak's manpower had reached an apex of about 145,000 in the 1980s. By 2012, layoffs since 2003 had slashed 47,000 jobs, alongside the shutdown of 13 factories and some 130 labs.

On 19 January 2012 the 131-year-old firm sought Chapter 11 protection from its creditors, sapped by pension costs and years of slowness in going digital. Citigroup, a lender, provided $950 million for its restructuring. A portfolio of 1,100 digital imaging patents fetched about $525 million from a consortium of twelve buyers, including the giants Apple and Google.

After 19 months of reorganisation, the once-mighty Kodak left bankruptcy in September 2013, downsized, streamlined, and refocused as a commercial printer for business clients. In November 2009 President Obama had decorated Sasson with national laurels, commending his digital camera. That medal's year also saw Kodak announce the end of Kodachrome.

中文译文

1975年，柯达工程师史蒂文·萨松造出第一台数码相机。这台烤面包机大小的装置是个约八磅重的笨重大块头，几乎谈不上便携，更不用说手持。它的第一张粗糙照片只有一万个黑白像素；把照片记录到盒式磁带上需要23秒，再由回放装置显示在电视上。

柯达的营收显示，公司依赖胶卷这项利润支柱。技术人员很欣赏这台相机，可它不需要胶卷。据萨松回忆，总部上司说它挺可爱，却让他别声张。1978年，相机取得专利，但这台原型机始终没有投产。

数码摄影的问世带来颠覆性冲击。2009年6月，柯达宣布将停产Kodachrome这一传统胶卷产品，理由是数码时代需求减少。公司员工规模曾在20世纪80年代达到约14.5万人的顶峰。到2012年，自2003年起的裁员已削减4.7万个岗位，同时关闭了13家工厂与约130个冲印实验室。

养老金成本与多年数码转型迟缓不断削弱公司。2012年1月19日，这家拥有131年历史的企业申请美国《破产法》第11章保护，以免债权人追讨。贷款方花旗集团提供9.5亿美元用于重组。包括苹果、谷歌等巨头在内的十二家买方组成财团，以约5.25亿美元买下1100项数字成像专利组成的专利组合。

经过19个月重组，昔日强大的柯达于2013年9月走出破产程序：缩小规模、精简业务，重新聚焦于为企业客户提供商业印刷服务。早在2009年11月，奥巴马总统就授予萨松国家级荣誉，表彰他的数码相机。而在这枚奖章颁发的同一年，柯达也宣布了Kodachrome胶卷的终结。

史实说明：这是2009年度的国家技术与创新奖章，实际颁发日为2010年11月17日（白宫典礼记录）。柯达宣布停产Kodachrome是在2009年，因此两件事并非发生于同一年。

核心词汇

先记这22项。连着下方的原文短语一起回想。

cumbersome adj.
笨重的
a cumbersome heavyweight

rudimentary adj.
初步的；粗陋的
Its first rudimentary picture

revenue n.
营收
Kodak's revenues

dependence on 名词短语
对……的依赖
its dependence on film

profitable adj.
盈利的；赚钱的
the profitable pillar of its business

appreciative of 形容词短语
欣赏；赞赏
appreciative of the camera

recollection n.
回忆
By Sasson's recollection

keep … hushed 动词短语
对此保密；别声张
urged him to keep it hushed

patent n.
专利
A patent followed in 1978

advent n.
出现；问世
The advent of digital photography

disruptive adj.
颠覆性的
proved disruptive

layoff n.
裁员
layoffs since 2003

creditor n.
债权人
protection from its creditors

sap v.
逐渐削弱
sapped by pension costs

restructuring n.
重组
for its restructuring

portfolio n.
一组持有的资产；此指专利组合
A portfolio of 1,100

consortium n.
财团；联合体
a consortium of twelve buyers

reorganisation n.
重组；改组
19 months of reorganisation

downsize v.
缩小规模
downsized, streamlined

streamline v.
精简
downsized, streamlined

refocus as 动词短语
重新定位为
refocused as a commercial printer

commend v.
表彰
commending his digital camera

文中词汇与搭配

补充词汇共33项。保留每个表达在本文中的意思。

digital camera 名词短语
数码相机

toaster n.
烤面包机

gadget n.
小装置

heavyweight n.
重家伙；大块头

portable adj.
便携的

let alone 固定短语
更不用说

handheld adj.
手持的

pixel n.
像素

cassette n.
盒式磁带

playback n.
回放

pillar n.
支柱

staffer n.
员工

superior n.
上司

headquarters n.
总部

analogue adj.
模拟式的；此处指传统胶片

lessen v.
减少

era n.
时代

manpower n.
员工规模；人力

apex n.
顶峰

slash v.
大幅削减

shutdown n.
关闭

pension n.
养老金

slowness n.
迟缓

lender n.
贷款方

buyer n.
买方

giant n.
行业巨头

once-mighty adj.
昔日强大的

printer n.
印刷企业；不是打印机

client n.
客户

decorate v.
授予荣誉；授勋

laurels n.
荣誉

medal n.
奖章

the end of 名词短语
……的终结

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 The _____ of digital photography proved disruptive.

02 By Sasson's _____, his superiors at headquarters called it cute and urged him to keep it hushed.

03 Citigroup, a lender, provided $950 million for its _____.

04 downsized, _____, and refocused as a commercial printer for business clients.

文章主线

- 先机 1975年，柯达工程师萨松造出第一台数码相机：约八磅重、一万黑白像素，存一张照片需23秒。
- 牵制 胶卷是利润支柱；据萨松回忆，上司让他别声张。原型机1978年获专利，却未投产。
- 衰退 数码摄影冲击胶卷需求，转型迟缓与养老金负担削弱公司，柯达于2012年申请破产保护。
- 重组 柯达获9.5亿美元贷款，并以约5.25亿美元出售1100项专利；2013年完成重组，转做企业印刷。
- 反差 柯达2009年宣布停产Kodachrome胶卷；萨松获2009年度美国国家技术与创新奖章，实际于2010年授勋。

参考答案　1 advent 2 recollection 3 restructuring 4 streamlined

青蒿一握，济世千万

A Handful of Wormwood That Saved Millions



古方里没有煮沸这一步，怎样启发了青蒿素研究？

READ 纯净阅读 | CHECK 中文核对 | LEARN 词汇记忆 | RECALL 回想巩固

英文原文

Project 523, a covert military mission involving many institutes, was named after its launch date, 23 May 1967. Its urgent target was the chloroquine-resistant malaria weakening troops in the Vietnam War. In January 1969 Tu Youyou was entrusted with leading her institute's research group, despite having no doctorate or foreign training.

In an exhaustive search, her team combed over 2,000 folk recipes, compiled 640 prescriptions, and evaluated about 380 extracts from some 200 medicinal herbs. Qinghao, or sweet wormwood, showed flashes of promise, yet its results were erratic. Seeking a clue, Tu revisited the classical texts.

Around AD 340, the physician Ge Hong's handbook advised taking "a handful of qinghao immersed in two litres of water". This bypassed boiling. Tu suspected heat during extraction was degrading a delicate component, so she switched to ether, a volatile solvent boiling at about 35 degrees Celsius. On 4 October 1971 sample 191 completely inhibited the parasite in infected mice.

Efficacy in mice indicated nothing about human safety. In July 1972, to assess its toxicity, Tu and two colleagues voluntarily took the extract. Thereafter, from August to October, it was administered to 21 malaria patients in Hainan, and on 8 November her team obtained pure crystals of the substance later named artemisinin.

In September 2011 Tu was named a Lasker Award recipient; its jury credited the drug with saving millions. On 5 October 2015 she won half the Nobel Prize in Physiology or Medicine, the first Chinese woman Nobel laureate. Since 2006 WHO guidelines have endorsed artemisinin-based combination therapies against malaria's deadliest form. The essence of Ge Hong's line lay in the boiling it skipped.

中文译文

“523项目”是一项多家研究机构参与的秘密军事任务，名称来自启动日期：1967年5月23日。项目急需解决的是对氯喹产生耐药性的疟疾，当时这种疾病正削弱越南战争中的部队。1969年1月，尽管没有博士学位，也未在国外受过训练，屠呦呦仍受命领导所在研究所的研究组。

团队进行了全面搜寻，仔细翻检2000多份民间药方，汇编640个方剂，评估了来自约200种药用植物的约380种提取物。青蒿，也就是黄花蒿，时而显出效果，结果却不稳定。为寻找线索，屠呦呦重新翻阅古代文献。

公元340年前后，医师葛洪的手册建议取“一把浸在两升水中的青蒿”。这种做法不经煮沸。屠呦呦怀疑，提取过程中的高温会破坏某种不耐热的成分，于是改用乙醚：一种沸点约35摄氏度、容易挥发的溶剂。1971年10月4日，第191号样品完全抑制了感染小鼠体内的疟原虫。

小鼠身上有效，并不能说明对人体是否安全。1972年7月，为评估毒性，屠呦呦与两名同事自愿服下提取物。随后，8月至10月间，提取物用于海南21名疟疾患者；11月8日，团队取得了这种物质的纯净晶体，它后来被命名为青蒿素。

2011年9月，屠呦呦获拉斯克奖，评审团认为这种药物挽救了数百万人的生命。2015年10月5日，她获得当年诺贝尔生理学或医学奖的一半份额，成为首位中国女性诺贝尔奖得主。自2006年起，世卫组织指南就推荐以青蒿素为基础的多药联合疗法，对抗最致命的一类疟疾。葛洪那句话的关键，恰在于没有煮沸。

核心词汇

先记这22项。连着下方的原文短语一起回想。

resistant adj.
有抗药性的
chloroquine-resistant malaria

be entrusted with 动词短语
受托承担；受命负责
entrusted with leading

exhaustive adj.
全面彻底的
In an exhaustive search

comb v.
仔细搜寻；翻检
combed over 2,000 folk recipes

compile v.
汇编
compiled 640 prescriptions

evaluate v.
评估
evaluated about 380 extracts

extract n.
提取物
evaluated about 380 extracts

erratic adj.
不稳定的；时好时坏的
its results were erratic

extraction n.
提取
heat during extraction

degrade v.
降解；使成分受破坏
was degrading a delicate component

component n.
成分
a delicate component

volatile adj.
易挥发的
a volatile solvent

solvent n.
溶剂
a volatile solvent

inhibit v.
抑制
completely inhibited the parasite

efficacy n.
药效；效力
Efficacy in mice

assess v.
评估
to assess its toxicity

toxicity n.
毒性
to assess its toxicity

administer … to … 动词短语
给……用药
administered to 21 malaria patients

substance n.
物质
pure crystals of the substance

endorse v.
认可；推荐
have endorsed artemisinin-based

combination n.
联合；合用
combination therapies

essence n.
关键；精髓
The essence of Ge Hong's line

文中词汇与搭配

补充词汇共44项。保留每个表达在本文中的意思。

covert adj.
秘密的；隐蔽的

military adj.
军事的

mission n.
任务

urgent adj.
紧迫的

malaria n.
疟疾

weaken v.
削弱

troop n.
部队

Vietnam War 专名
越南战争

despite prep.
尽管

doctorate n.
博士学位

folk adj.
民间的

prescription n.
药方

medicinal adj.
药用的

herb n.
草药；药用植物

sweet wormwood 名词短语
黄花蒿

flash n.
短暂显现

promise n.
有效的希望；前景

clue n.
线索

revisit v.
重新审阅

classical adj.
古代经典的

handbook n.
手册

be immersed in 动词短语
浸在……中

文中词汇与搭配 续

补充词汇共44项。保留每个表达在本文中的意思。

litre n.
升

bypass v.
避开；略过

delicate adj.
易受损的；此处指不耐热的

switch to 动词短语
改用

ether n.
乙醚

degree Celsius 名词短语
摄氏度

parasite n.
寄生虫；此处指疟原虫

infected adj.
受感染的

indicate v.
表明；说明

voluntarily adv.
自愿地

thereafter adv.
此后

patient n.
患者

obtain v.
获得

crystal n.
晶体

recipient n.
获奖者

jury n.
评审团

physiology n.
生理学

laureate n.
获奖者；此处指诺奖得主

guideline n.
指南

therapy n.
疗法

deadly adj.
最致命的

skip v.
略过；省去

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 This _____ boiling.

02 Efficacy in mice indicated nothing about human _____.

03 In July 1972, to assess its _____, Tu and two colleagues voluntarily took the extract.

04 The _____ of Ge Hong's line lay in the boiling it skipped.

文章主线

- 任务 523项目联合攻关，目标是对旧药氯喹有抗药性的疟疾；屠呦呦带队查2000多药方，试约380种提取物。
- 线索 青蒿药效不稳；葛洪古方浸泡青蒿而不煮沸，启发屠呦呦怀疑高温会破坏有效成分。
- 突破 改用沸点约35°C的乙醚低温提取，1971年第191号样品完全抑制小鼠疟原虫；人体安全仍待验证。
- 验证 1972年，她与两名同事试服提取物，再用于21名疟疾患者，随后得到青蒿素晶体。
- 影响 以青蒿素为基础的多药合用疗法获世卫推荐，挽救众多生命；屠呦呦2015年获诺奖。

参考答案　1 bypassed 2 safety 3 toxicity 4 essence

洞穴里的十三个人

Thirteen in the Cave



找到十三人之后，为什么救他们出洞仍如此困难？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On 23 June 2018, twelve boys aged 11 to 16 from a junior football squad went on an outing with their 25-year-old assistant coach. They entered Tham Luang, a sprawling maze of tunnels in northern Thailand, to penetrate its gloom. Monsoon rain then flooded the passages, and the rising water trapped all of them.

For nine days their whereabouts were unknown. On 2 July two elite British divers, threading through submerged passages, reached an inundated chamber about 4 km from the entrance. All 13 were there, huddled on a muddy mound and perched just above the water, every one of them alive.

The only exit ran through kilometres of constricted, treacherous passages, many of them navigable only with scuba gear. Rescuers began stockpiling oxygen cylinders along the route. On 6 July Saman Kunan, a former Thai Navy SEAL, fell unconscious while laying them and died.

The evacuation began on 8 July, and the boys came out in batches. Each was given doses of sedative, wrapped motionless in a stretcher, and passed from diver to diver, with medics stationed along the route. On 10 July the last four footballers and their coach were escorted out after their teammates.

The multinational mobilisation was a feat of logistics. Some 10,000 people converged on the site, including a contingent of more than 100 divers, plus 2,000 soldiers and 900 policemen, with teams deployed from abroad. The rescue had a sobering postscript: in December 2019 another Navy SEAL died of a blood infection caught during the operation. Every boy came home alive, and two rescuers were mourned.

中文译文

2018年6月23日，一支少年足球队的十二名男孩与25岁的助理教练外出游玩；孩子们年龄在11至16岁。他们进入泰国北部的睡美人洞，深入这个隧道交错、向四方延伸的幽暗迷宫。随后季风雨淹没通道，上涨的水将所有人困在洞内。

整整九天，没有人知道他们在哪里。7月2日，两名英国精英潜水员穿过淹没的通道，到达距入口约4公里的一处进水洞室。十三人全在那里，挤在一处刚高出水面的泥土堆上，每个人都还活着。

唯一的出路要经过数公里狭窄、暗藏危险的通道，其中许多路段只有借助水肺装备才能通行。救援人员开始沿途布放、储备氧气瓶。7月6日，前泰国海军海豹突击队员萨曼·古南在布放气瓶时失去知觉，随后死亡。

撤离于7月8日开始，男孩们分批出洞。他们分别接受镇静药，被固定包裹在担架中，保持不动，由潜水员逐段接力运送，沿途有医护人员驻守。7月10日，最后四名小球员与教练继其他队友之后，被护送出洞。

这场跨国动员堪称后勤协调的壮举。约一万人汇集现场，包括100多名潜水员组成的队伍，以及2000名士兵和900名警察，还有从国外调派来的救援团队。但救援还有一段令人沉痛的后续：2019年12月，另一名海军海豹突击队员死于救援行动中染上的血液感染。每个男孩都活着回家，两名救援者却长留在人们的悼念中。

核心词汇

先记这22项。连着下方的原文短语一起回想。

sprawling adj.
向四周延伸的
a sprawling maze of tunnels

maze n.
迷宫
a sprawling maze of tunnels

whereabouts n.
下落；所在之处
their whereabouts were unknown

thread through 动词短语
穿行于；小心穿过
threading through submerged passages

submerged adj.
淹没在水下的
submerged passages

inundated adj.
被水淹的
an inundated chamber

constricted adj.
狭窄的
constricted, treacherous passages

treacherous adj.
暗藏危险的
constricted, treacherous passages

navigable adj.
可以通行的
navigable only with scuba gear

stockpile v.
储备
stockpiling oxygen cylinders

unconscious adj.
失去知觉的
fell unconscious

evacuation n.
撤离
The evacuation began on 8 July

in batches 介词短语
分批地
the boys came out in batches

sedative n.
镇静药
doses of sedative

motionless adj.
一动不动的
wrapped motionless in a stretcher

escort v.
护送
were escorted out

multinational adj.
多国参与的
The multinational mobilisation

mobilisation n.
动员
The multinational mobilisation

logistics n.
后勤；运输与协调
a feat of logistics

deploy v.
调派；部署
teams deployed from abroad

sobering adj.
令人警醒或沉痛的
a sobering postscript

mourn v.
悼念
two rescuers were mourned

文中词汇与搭配

补充词汇共39项。保留每个表达在本文中的意思。

junior adj.
青少年的

squad n.
球队；小队

go on an outing 动词短语
外出游玩

assistant coach 名词短语
助理教练

enter v.
进入

tunnel n.
隧道

penetrate v.
深入；进入深处

gloom n.
幽暗

monsoon n.
季风

flood v.
淹没

passage n.
通道

elite adj.
精英的；顶尖的

diver n.
潜水员

entrance n.
入口

huddle v.
挤在一起

muddy adj.
泥泞的

mound n.
土堆

perch v.
栖身在狭小高处

alive adj.
活着的

exit n.
出口

文中词汇与搭配 续

补充词汇共39项。保留每个表达在本文中的意思。

run through 动词短语
经过；贯穿

kilometre n.
公里

scuba n.
水肺；自携式水下呼吸器

rescuer n.
救援人员

cylinder n.
气瓶

along the route 介词短语
沿途

former adj.
以前的；前任的

dose n.
剂量；一次用药量

wrap v.
包裹

stretcher n.
担架

medic n.
医护人员

station v.
派驻；安排驻守

footballer n.
足球队员

teammate n.
队友

converge on 动词短语
汇集到

contingent n.
分队；一支队伍

policeman n.
警察

postscript n.
后续；事后补记

blood infection 名词短语
血液感染

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 For nine days their _____ were unknown.

02 Rescuers began _____ oxygen cylinders along the route.

03 The evacuation began on 8 July, and the boys came out _____.

04 The multinational mobilisation was a feat of _____.

文章主线

- 被困 2018年，泰国12名少年球员和助理教练进入睡美人洞，因季风雨淹没通道而受困。
- 发现 失联九天后，两名英国潜水员在距洞口约4公里处找到他们，十三人全部活着。
- 难题 救援需穿过数公里狭窄水道；约万人的多国队伍沿途布放气瓶，协调百余名潜水员。
- 救出 7月8日至10日，男孩用镇静药后固定在担架中，由潜水员接力带出，沿途有医护协助。
- 代价 十三人全获救；救援者萨曼·古南在前期布气瓶时遇难，另一名救援队员2019年死于救援时染上的血液感染。

参考答案　1 whereabouts 2 stockpiling 3 in batches 4 logistics